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Multi-feature stock price prediction by LSTM networks based on VMD and TMFG

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The stock market is characterized by its high nonlinearity and complexity, making traditional methods ineffective in capturing its nonlinear features and complex market dynamics. This paper proposes a novel stock price forecasting model—the Variational Mode Decomposition—Triangulated Maximally Filtered Graph—Long Short-Term Memory (VMD–TMFG–LSTM) combined model—aimed at improving prediction accuracy, stability, and computational efficiency. The proposed model first employs Variational Mode Decomposition (VMD) to decompose the stock price time series into multiple smooth intrinsic mode functions (IMFs), reducing data complexity and mitigating noise interference. Subsequently, the TMFG algorithm is utilized for feature selection, simplifying the input data and accelerating the iterative convergence process. Finally, the filtered features are modeled and predicted using a Long Short-Term Memory (LSTM) network. Experimental results demonstrate that the VMD–TMFG–LSTM model significantly outperforms AutoRegressive Integrated Moving Average (ARIMA), Neural Network (NN), Deep Neural Network (DNN), Convolutional Neural Network (CNN), as well as single LSTM,

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data through its original gating mechanism, which enables LSTM to identify and exploit long-term dependencies in the data, greatly enhancing the accuracy of the model for time series prediction.

Nonetheless, the high volatility, nonlinearity, chaos, and operational complexity of the stock market itself make accurately predicting stock prices still a great challenge. In order to effectively address these challenges, this study proposes a state-of-the-art combinatorial model (VMD–TMFG–LSTM) that incorporates variational modal decomposition (VMD), Triangulated Maximally Filtered Graph (TMFG), and LSTM network. This model starts with an accurate decomposition of stock closing price series using VMD technique to reduce the complexity of the data and to effectively remove unstructured noise. Immediately after that, the processed data is sparsely dimensionalised by the TMFG algorithm, a step that not only reduces the burden of data processing, but also highlights the core features that influence stock price movements. Finally, the filtered key features are fed into the LSTM network for

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reduces some of the test error noise interference and improves the prediction accuracy.

(3) The introduction of the TMFG algorithm to select the input features reduces the complexity of the input, improves the speed of iterative convergence and reduces the iterative error, and improves the stability and robustness of the prediction.

The rest of the paper is organised as follows, the second part is the literature review section, which provides an overview of existing stock prediction models and methods, discusses data decomposition and feature selection techniques, and introduces the application of TMFG algorithm to filtered data and the application of LSTM to stock prediction. The third part is the Methods and Data section, which details the principles of VMD decomposition, the TMFG algorithm and the LSTM network, and describes the construction of the combined VMD–TMFG–LSTM model. It also describes the data sources and data preprocessing for this study. The fourth part is the experimental and result analysis section, which demonstrates the experimental process and results of VMD decomposition, TMFG processing and LSTM

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Vector Machines (SVM) [7] and Gradient Boosting (Boost) [8] have shown good results in stock price prediction. These methods are better suited for capturing complex nonlinear relationships, particularly in financial time series data, which often exhibit such characteristics. As technology advances, especially in the area of neural networks, deep neural networks have gained widespread attention in the analysis and prediction of stock price fluctuations. Models such as Convolutional Neural Networks (CNN) [9] and Long Short-Term Memory (LSTM) [10] have become particularly prominent. Researchers have developed numerous improved algorithms and optimization strategies to enhance prediction accuracy, achieving significant results in practical applications. These innovations not only advance financial forecasting technology but also provide a more scientific foundation for investment decision-making.

As a typical temporal prediction network in deep learning, the LSTM neural network has performed well in the field of stock price forecasting. For instance, Sun et al. [10] used an

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VMD–LSTM model for time-varying series forecasting, demonstrating that VMD–LSTM can better fit time-varying data compared to EMD and single LSTM networks.

In addition to data decomposition, researchers have addressed the challenges of high volatility, nonlinearity, and chaos in the stock market by focusing on feature selection and data dimensionality reduction. Li et al. [16] applied the Least Absolute Shrinkage and Selection Operator (LASSO) to reduce the dimensionality of basic trading, valuation, and technical indicators, and then combined the processed data with an LSTM network for prediction. Their results showed that the LSTM model, after dimensionality reduction, achieved superior prediction accuracy. Similarly, Chen et al. [17] proposed a multi-feature stock price prediction model based on multi-feature computation, LASSO feature selection, and Ca-LSTM networks, considering the importance of each feature. Their experiments demonstrated that this model outperformed traditional time-series prediction models and standard LSTM methods in terms of both accuracy and speed.

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computationally faster and more suitable for large datasets [22]. Briola and Aste [23] study utilized Minimum Spanning Tree (MST) and Triangulated Maximally Filtered Graph (TMFG) methods to analyze the logarithmic return cross-correlations of 25 cryptocurrencies traded on the FTX exchange across different time resolutions (ranging from 15 s to 1 day), uncovering the evolution of dynamic hierarchical structures over varying time scales. The study demonstrated that TMFG effectively captures key dependency structures across time resolutions with stability, statistical significance, and economic relevance. This provides significant support for the VMD–TMFG–LSTM model in stock prediction, particularly in feature selection in stock markets. Similarly, Hussain Shahzad et al. [24] applied the TMFG algorithm to construct an energy stock return dependence network using data from 105 global energy companies across 26 countries in the Americas, Asia–Pacific, and Europe. The extracted significant dependencies facilitated the development of a portfolio trading strategy, which outperformed the traditional equal-weighted buy-and-hold approach. These findings support the applicability of the TMFG

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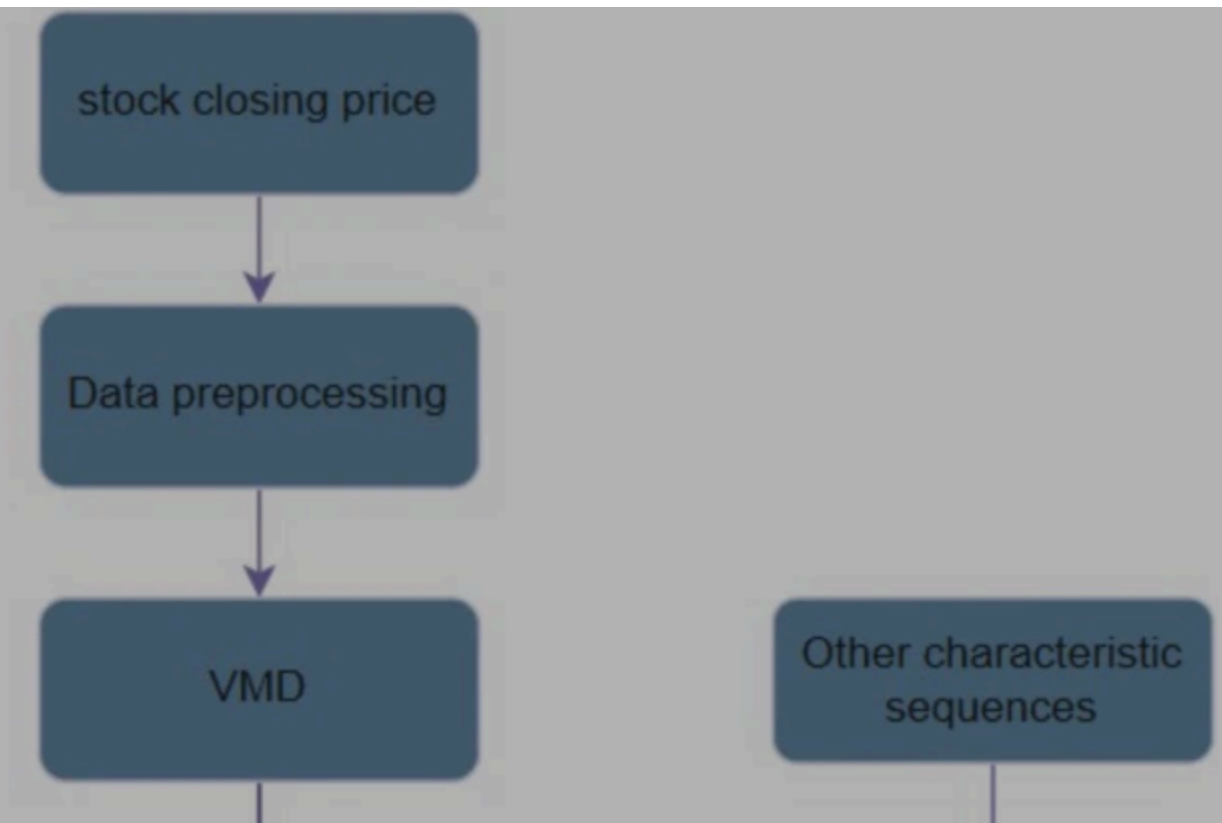
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The overall process consists of five main components:

- (1) Obtain the price-related information of the target stock, including its opening price, closing price, high price, low price, Changes, volume, turnover, Circulation market value, total market value, turnover rate, restoration price, P/E ratio, P/S ratio, P/C ratio, P/B ratio, closing price recovery price, and other 16 indicators;
- (2) Calculating the technical indicators from the original stock data resulted in 43 stock technical indicators, which were merged into the original data to obtain new experimental data.
- (3) To process the experimental data, in order to better retain the characteristics of the original data, we use the VMD variational model decomposition technique to process

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Variational modal decomposition (VMD) is a nonlinear, nonsmooth signal analysis method for signal processing [26]. It decomposes a complex signal into a series of intrinsic modal functions (IMFs) through adaptive decomposition, where each IMF represents an oscillatory mode in the signal [27]. Compared with the classical empirical modal decomposition (EMD), VMD avoids the phenomenon of modal aliasing by introducing constraints, and VMD has better stability and accuracy in processing signals, which is especially suitable for dealing with multi-frequency signals.

The VMD decomposition is given by:

$$s_k(t) = \left(\delta(t) + \frac{j}{\pi t} \right) u_k(t)$$

(1)

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graph, thus ensuring the sparsity of the network while maintaining a high degree of structure. TMFG is used to deal with the structure of the clusters in graphical models, and is particularly suited to time-series analyses in the financial domain, e.g., modelling of stock returns. By calculating the maximal clusters, it is possible to identify the set of variables that are interrelated in a time series, which can then be used for risk management and portfolio optimisation. It can effectively reveal the complex interactions between variables. The TMFG algorithm can be used to analyse stock datasets by calculating the clusters structure to identify the strong correlations between variables and screen the features for stock price prediction. The ‘sparsity’ of the sparse inverse covariance matrix means that most of the elements of the matrix are zero, which is very useful in practical applications because it reflects only the key direct relationships between the variables and ignores indirect or non-significant relationships. This property makes the model easier to interpret and significantly reduces computational complexity and improves computational efficiency when dealing with high dimensional data.

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where $\{f_{C_m}(X_{C_m})\}$ is the local distribution of the clusters $\{C_m\}$, $\{f_{S_n}(X_{S_n})\}$ is the local distribution of the separations $\{S_n\}$, and $k(\{S_n\})$ is the normalisation constant of the separations $\{S_n\}$. Similarly, J can be decomposed, and we can obtain J from the summation of the local inverse of the correlation matrix.

$$J_{i,j} = \sum_{C: i,j \in C} \left(\text{vec}(\Sigma_C)^{-1} \right)_{i,j} - \sum_{S: i,j \in S} (k(S)-1) (\text{vec}(\Sigma_S)^{-1})_{i,j} \quad (5)$$

where $J_{i,j}$ is the covariance matrix element between variables i, j , $\left(\text{vec}(\Sigma_C)^{-1} \right)$ is the inverse covariance matrix of regiment C , and $k(S)$

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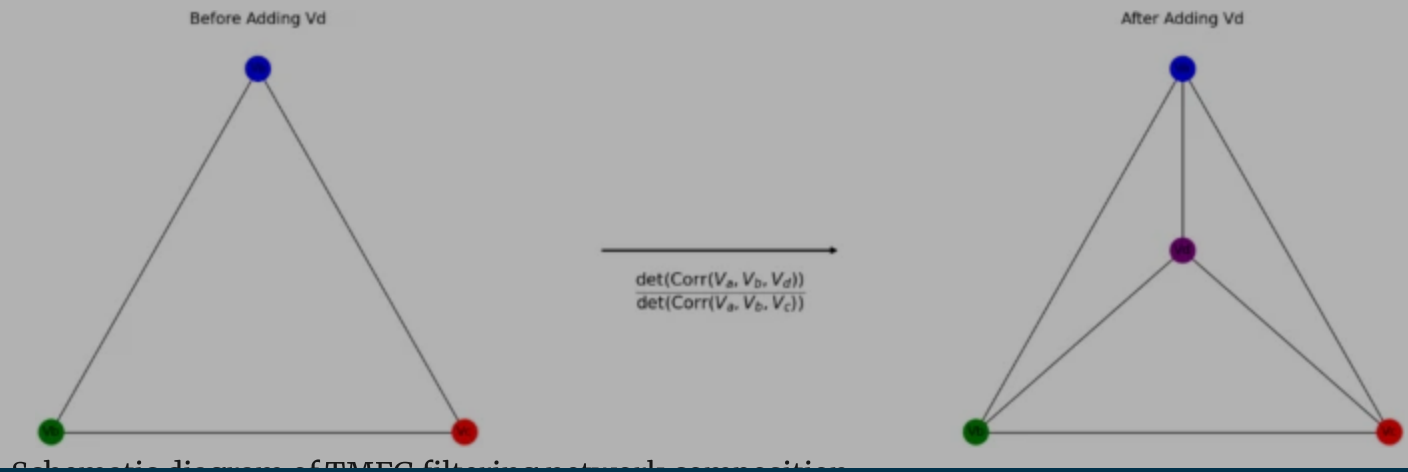
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returned with non-zero elements indicating the existence of edges (direct relationships) between variables in the TMFG network and zero elements indicating that the two variables are conditionally independent.

Fig. 2



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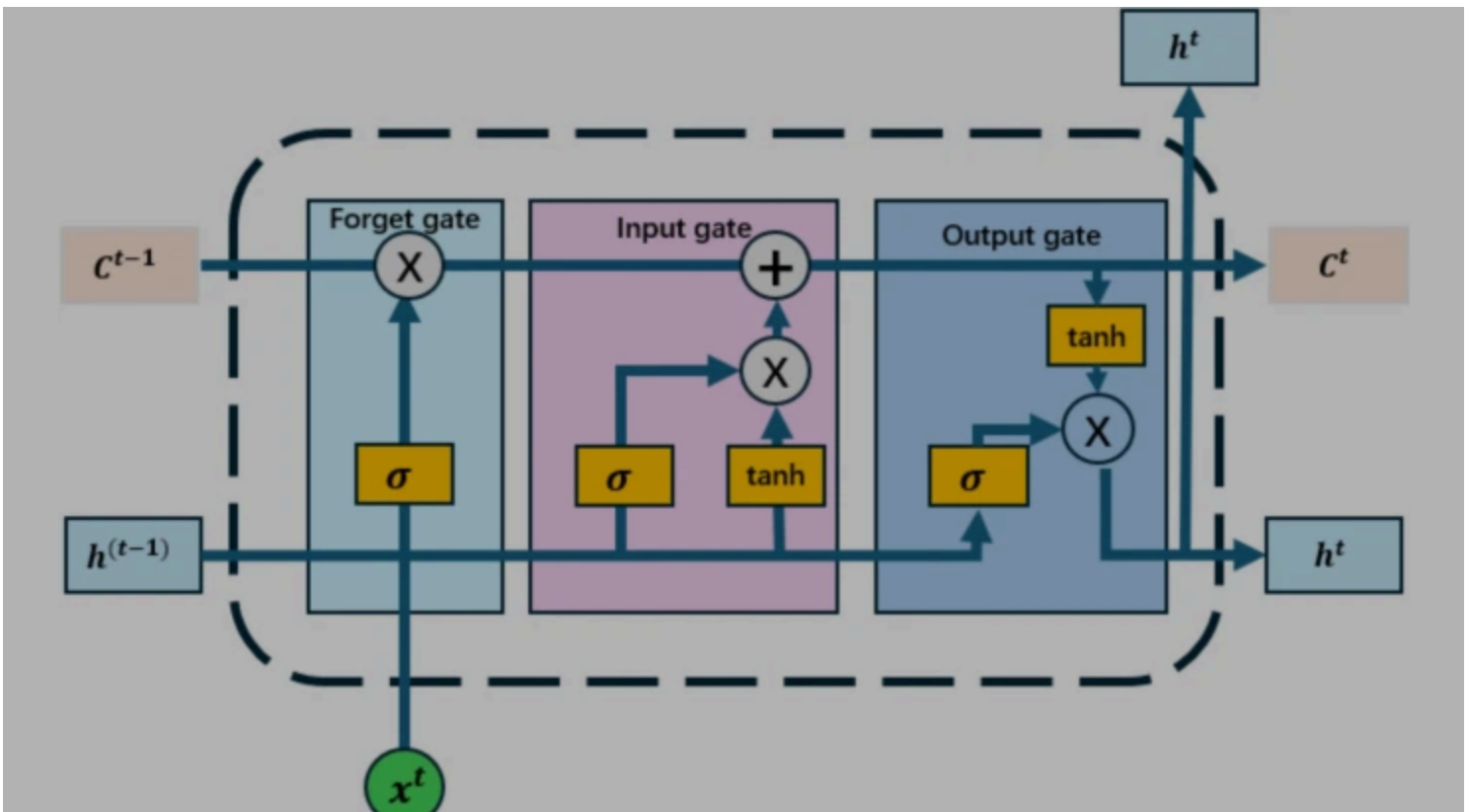
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(9)

$$\begin{array}{c} c^t = \sigma(W_o[h^{(t-1)}, x^t] + b_o) \\ o^t = \tanh(c^t) \end{array}$$

(10)

where (W_i, W_c, W_o) represent the corresponding weights, (b_i, b_c, b_o) represent the corresponding biases, and c^t represents the current cell state value, and h^t is the output value of the current cell.

Evaluation indicators

In this paper, the commonly used method of evaluating regression prediction algorithms is used to evaluate the model effect, and four evaluation indexes, namely, Root Mean Square

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usually ranges from 0 to 1. The closer the value is to 1, the better the predictive ability of the model and the more variability it explains.

The formula is:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \widehat{y}_i)^2}{\sum_{i=1}^n (y_i - \overline{y})^2} \quad (13)$$

where (y_i) is the actual value, $(\widehat{y}_i)$ is the predicted value, $(\overline{y})$ is the average of the actual values, and n is the number of data points.

The model run time formula is used to measure the efficiency of the model run, by

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value, total market value, turnover rate, restoration price, P/E ratio, P/S ratio, P/C ratio, P/B ratio, closing price recovery price are the original data. The original data indicators are shown in Table 1.

Table 1 16 Raw data indicators

When constructing predictive models, the use of multiple price-related variables provides a comprehensive view of the market. Although the primary focus in this paper is on the closing price for prediction, incorporating additional price-related variables such as opening price, high price, low price, and trading volume enhances the model's ability to capture market conditions. These variables complement the closing price by providing insights into

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[31]. In practice, technical indicators are used as features input into a machine learning model, which learns the historical patterns of these technical indicators to predict future stock price movements, in addition, the use of technical indicators can help overcome the limitations of traditional fundamental analysis as they are not influenced by the fundamental factors of the company, but rather focus on the stock price movement itself. When selecting technical indicators, it is necessary to consider their relevance, stability and diversity with respect to stock price movements. By carefully selecting and processing these technical indicators, powerful predictive models can be constructed to provide strong support for stock market analyses and decision-making. In this paper, we have selected several commonly used stock price technical indicators. The specific technical specifications are shown in Table 2.

Table 2 43 Stock technical indicators

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where max is the maximum value of the sample data, min is the minimum value of the sample data, and x is the original value to be normalized.

Experiments and analysis of results

VMD variational modal decomposition

VMD (Variational Mode Decomposition) is a signal processing technique that is commonly used to decompose time series data in order to extract its intrinsic patterns and trends [32, 33]. Compared with traditional smoothing techniques, VMD better preserves the characteristics of the original data during the decomposition process and therefore is more likely to capture important information and trends in stock price data. We finally set the parameters of VMD decomposition as follows: alpha (bandwidth constraint) is set to 7000, tau (noise tolerance) is set to 0, the DC value is 0, the init (initialisation mode, 1 means

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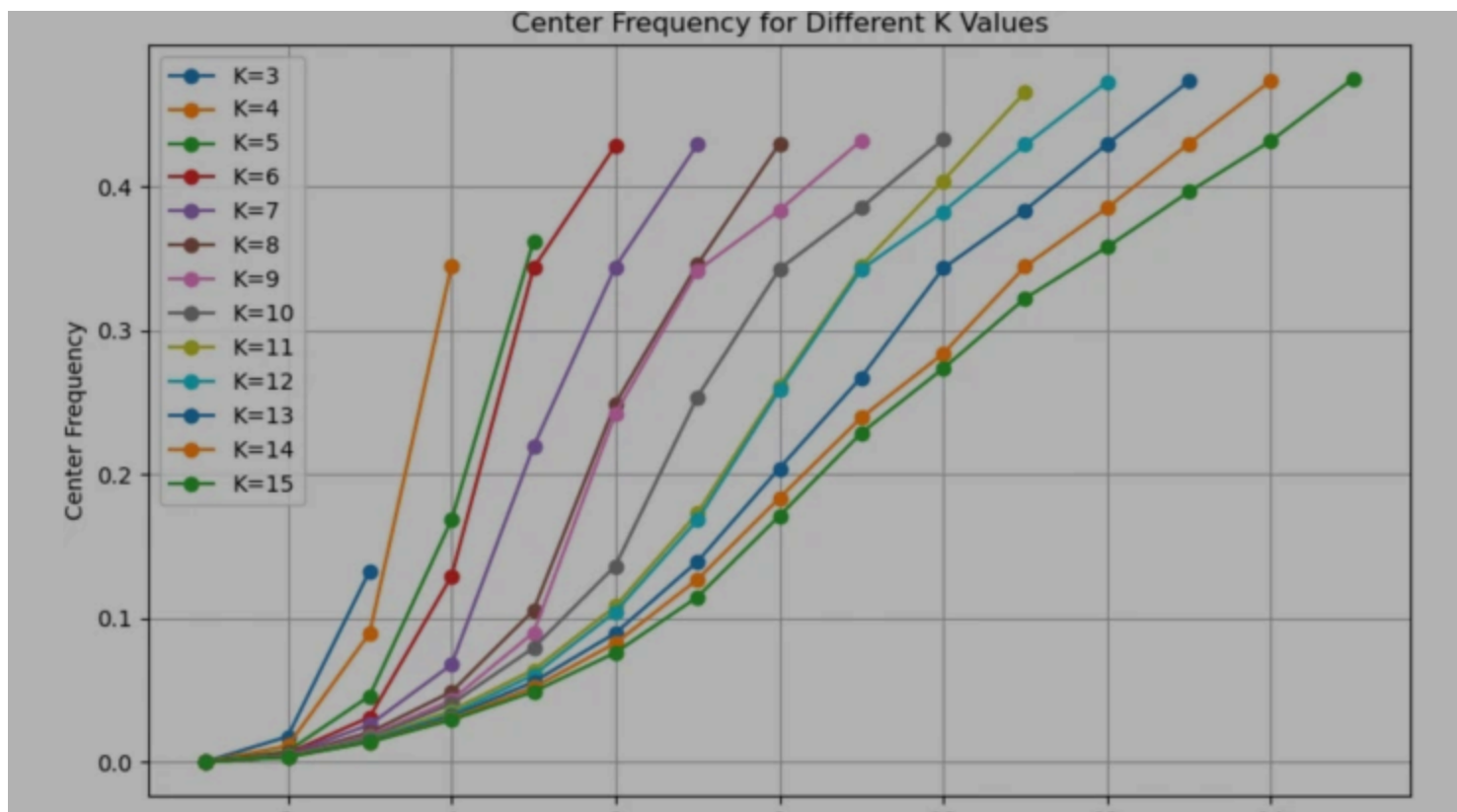
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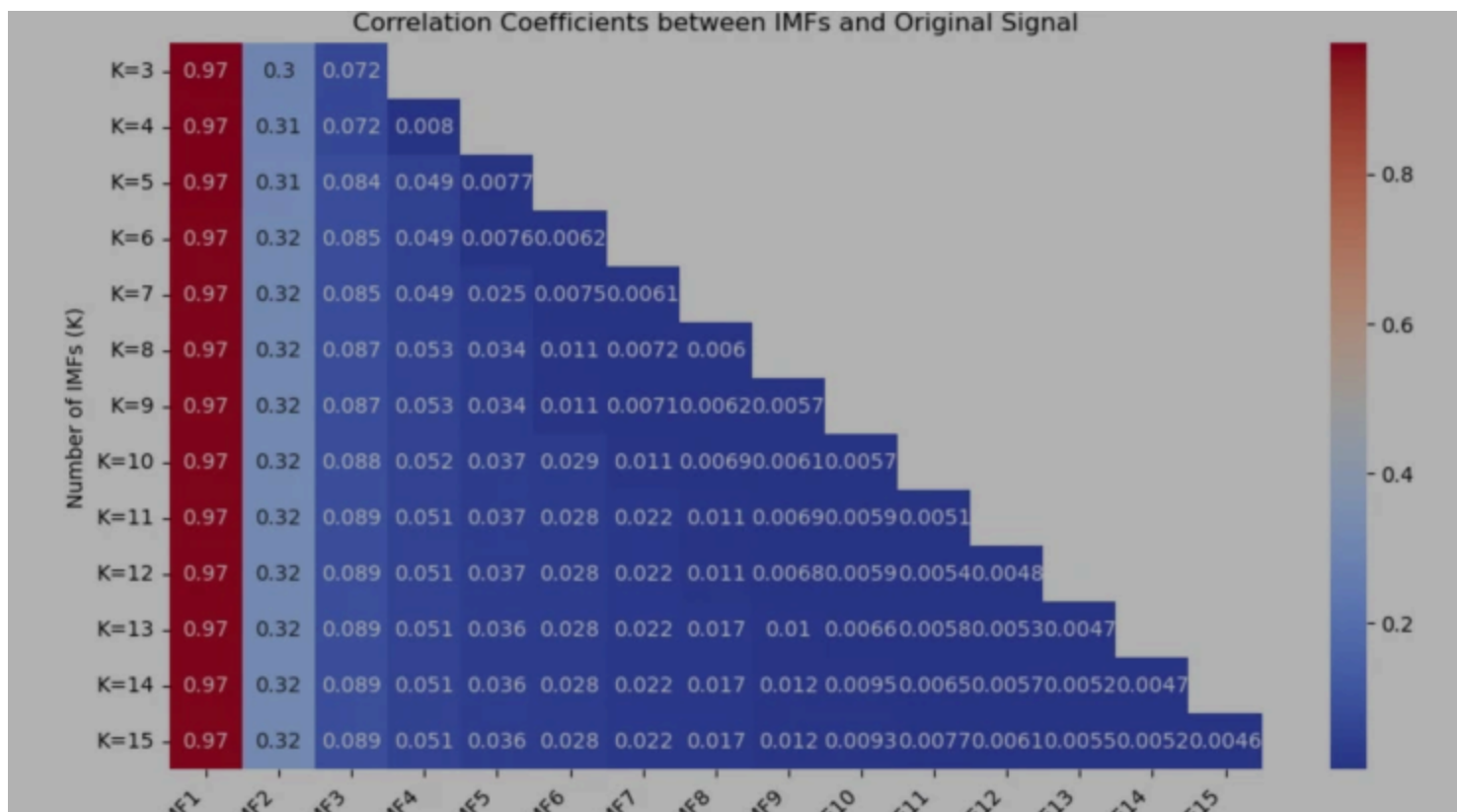
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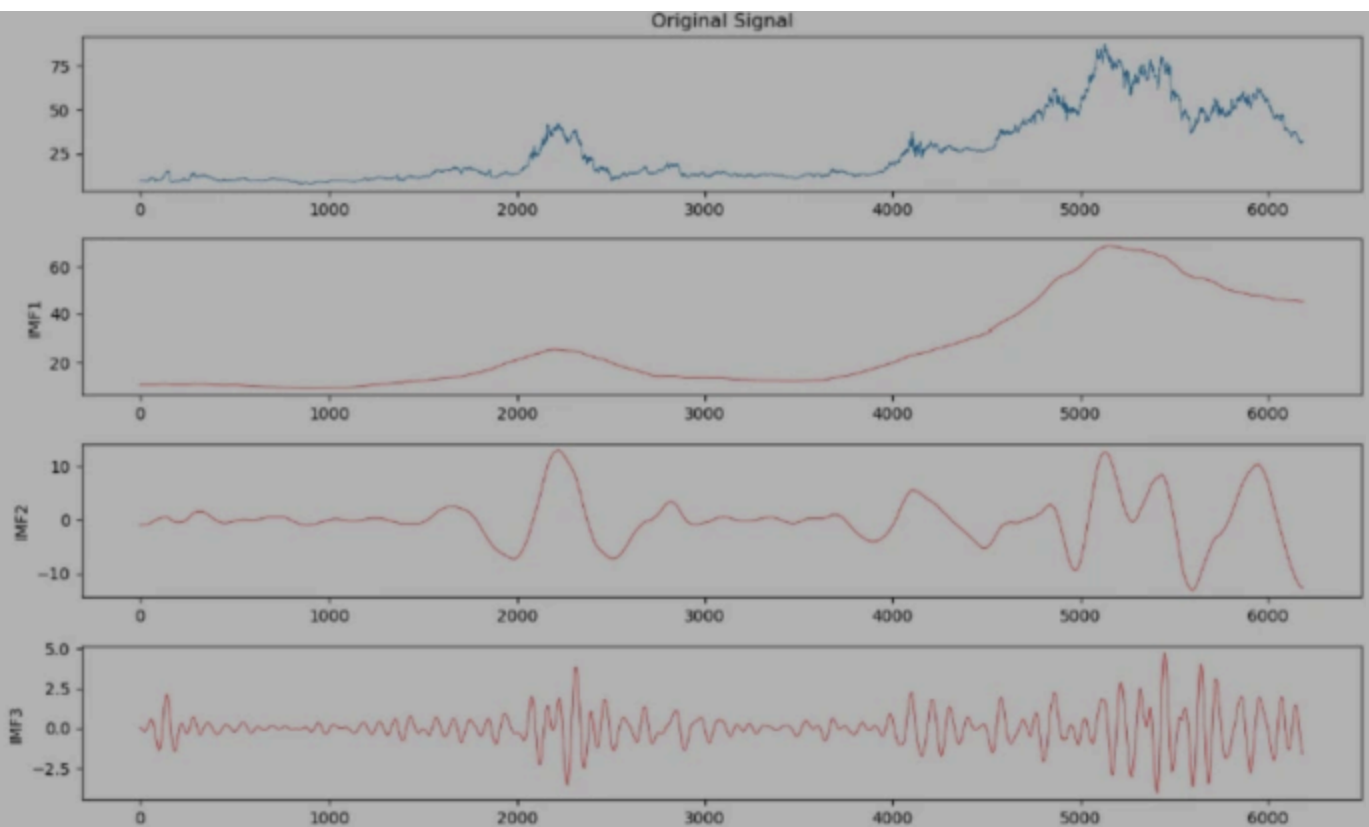
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As demonstrated in Fig. 6, the IMF1 component captures the low-frequency trend in the data, reflecting the overall upward and downward trend with smooth variations. The IMF2 component reflects the lower-frequency fluctuations, showing larger periodic fluctuations. The IMF3 component captures the mid-frequency fluctuations, containing more details and oscillations. The IMF4 component shows the higher-frequency fluctuations, which are more fine-grained and contain more high-frequency noise and detail. The IMF5-IMF8 components capture higher-frequency detail and show more violent and rapid fluctuations. These components typically contain more noise and irregular fluctuations, reflecting the high-frequency component of the market.

TMFG processing with LSTM prediction

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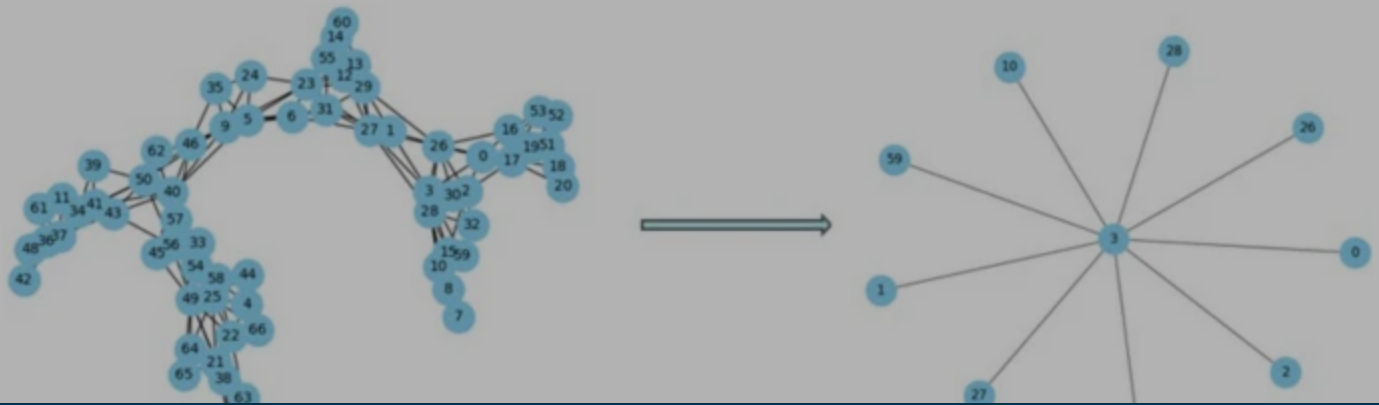
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among the complex feature relationships of 67 indicator data of Shanghai Airport (sh600009) stock, we extracted Open (node 0), High (node 1), Low (node 2), Close (node 3), adjust_price (node 10), SMA_20 (node 15), PP (node 26), R1 (node 27), S1 (node 28), and IMF_1 (node 59) total 10 key features.

Fig. 7



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In the task of stock price prediction, the proposed VMD–TMFG–LSTM model demonstrates superior predictive performance compared to other benchmark models (ARIMA, NN, DNN, CNN, LSTM) and ablation models (VMD–LSTM, TMFG–LSTM). The ARIMA model performs poorly in capturing nonlinear relationships, with its predicted values tending to flatten, especially during periods of sharp stock price fluctuations, failing to fit the actual trends. While the CNN model shows improvement in extracting local features, its performance is still limited due to its inability to model the long-term dependencies inherent in time series data, resulting in significant prediction errors. The DNN model, although advantageous in processing high-dimensional data, similarly lacks time-series modeling capabilities, leading to suboptimal fitting results for time-series data. The LSTM model, as the baseline, can partially capture time-series trends but exhibits limited fitting capability during rapid stock price changes. Through ablation analysis, the TMFG–LSTM model effectively reduces data redundancy and noise using the TMFG algorithm, significantly improving predictive stability in highly volatile regions. On the other hand, the VMD–LSTM model incorporates

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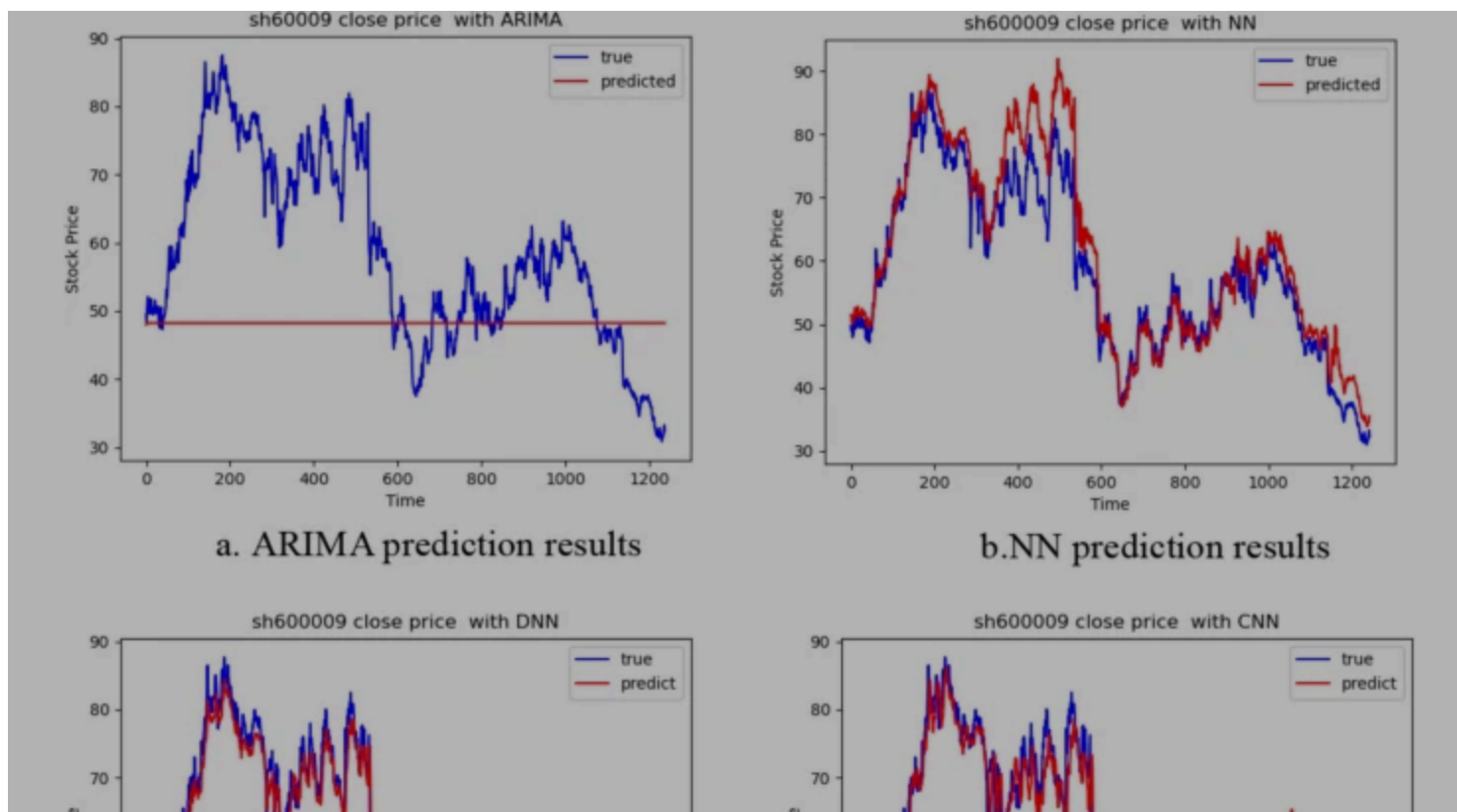
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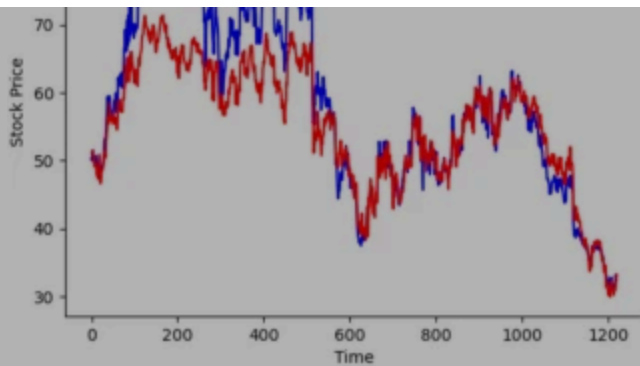
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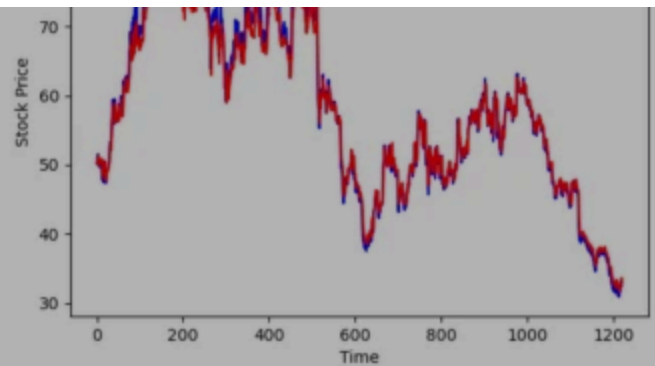
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g. VMD-LSTM prediction results



h. VMD-TMFG-LSTM prediction results

Shanghai International Airport Co., Ltd. (sh600009) Stock Price Forecast Results

In order to improve the accuracy and credibility of the prediction comparisons among eight models, four evaluation metrics were adopted, namely Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination (R-Square, R^2), and Model

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Table 3 Test set performance indicators of the eight models in Shanghai International Airport Co., Ltd. (sh600009) stock price forecasting

Validation of model validity

To validate the effectiveness and robustness of the proposed model, four SSE stocks were selected for experimental verification: China Resources Double-Crane Pharmaceutical Co., Ltd. (sh600062), China Meheco Group Co., Ltd. (sh600056), China Minsheng Banking Corp., Ltd. (sh600016), and Shanghai Waigaoqiao Free Trade Zone Group Co., Ltd. (sh600648). The experimental setup, including the environment, data sources, data preprocessing methods, applied models, and model parameters, is consistent with those used in the experiment for Shanghai International Airport Co., Ltd. (sh600009).

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Table 4 Test set performance metrics for stock price prediction of four stocks

Conclusion

To improve the stability and accuracy of stock price prediction and enhance the computational efficiency of the model, this paper integrates the VMD algorithm, the triangular maximal filtering network (TMFG) algorithm, and the LSTM network model, proposing a combined model—VMD–TMFG–LSTM. Based on closing price prediction experiments conducted on five stocks from the Shanghai Stock Exchange, the proposed model outperforms seven other models with smaller RMSE, R^2 , and MAE values and the fastest running time. In complex stock price prediction tasks, the TMFG method plays a

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The data in this paper are available from the website yucezhe (<https://yucezhe.com/>).

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Author information

Authors and Affiliations

College of Mathematics and Computer Science, Zhejiang A & F University, Hangzhou, 311300, China

Zhixin Zhang, Yanrong Hu & Hongjiu Liu

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Zhixin Zhang, Qingyang Liu, Yanrong Hu and Hongjiu Liu declare that there is no conflict of interests regarding the publication of our paper “*Multi-feature stock price prediction by LSTM networks based on VMD and TMFG*”.

Disclosure

Qingyang Liu, Yanrong Hu and Hongjiu Liu are the joint first authors and co-corresponding authors.

Ethics approval and consent to participate

This article does not contain any studies with human participants or animals performed by any of the authors.

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